

Project Objective

Analyze solar energy adoption across India.

1. ☒ Identify states and cities leading in solar installation.
2. ☒ Compare solar brands based on cost, electricity generation, and profitability.
3. ☒ Evaluate the impact of government subsidies on solar investments.
4. ☒ Calculate investment recovery period (Payback Period).
5. ☒ Measure long-term financial benefits of solar energy.
6. ☒ Analyze environmental impact through CO₂ reduction and tree-equivalent benefits.
7. ☒ Demonstrate that solar energy is both a profitable investment and a sustainable energy solution.

Step 1: Data Loading

```
In [1]: import numpy as np
import pandas as pd
import matplotlib as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv(r"D:/OWN PROJECT/Self Learning Project/Solar Energy/Solar Energy")
df.head()
```

```
Out[2]:
```

	Customer_ID	Customer_Name	State	City	Customer_Type	Property_Type	Installation_
0	C0001	Client_1	Rajasthan	Jaipur	Industrial	Apartment	16/03
1	C0002	Client_2	Telangana	Hyderabad	Residential	Independent House	18/02
2	C0003	Client_3	Telangana	Hyderabad	Industrial	Factory	07/03
3	C0004	Client_4	Gujarat	Surat	Commercial	Warehouse	20/12
4	C0005	Client_5	Telangana	Hyderabad	Industrial	Factory	20/05

5 rows × 41 columns



Step 2: Data Understand

In [3]: `df.shape`

Out[3]: (4500, 41)

In [4]: `df.columns`

Out[4]: Index(['Customer_ID', 'Customer_Name', 'State', 'City', 'Customer_Type', 'Property_Type', 'Installation_Date', 'Solar_Brand', 'Solar_Capacity_KW', 'Number_of_Panels', 'Panel_Wattage', 'Inverter_Brand', 'Installation_Type', 'System_Type', 'Solar_Cost', 'Installation_Cost', 'Total_Project_Cost', 'Government_Subsidy', 'Net_Cost_After_Subsidy', 'Monthly_Units_Generated', 'Monthly_Units_Consumed', 'Monthly_Units_Exported', 'Monthly_Units_Saved', 'Annual_Units_Generated', 'Annual_Units_Consumed', 'Annual_Units_Saved', 'Electricity_Rate_Per_Unit', 'Monthly_Bill_Before_Solar', 'Monthly_Bill_After_Solar', 'Monthly_Savings', 'Annual_Savings', 'ROI_Years', 'Lifetime_Savings', 'Annual_Maintenance_Cost', 'Service_Warranty_Years', 'Panel_Warranty_Years', 'Performance_Warranty_Years', 'Expected_Lifespan_Years', 'CO2_Reduced_KG_Per_Year', 'Trees_Equivalent', 'Green_Energy_Percentage'], dtype='object')

In [5]: `df.describe()`

Out[5]:

	Solar_Capacity_KW	Number_of_Panels	Panel_Wattage	Solar_Cost	Installation_Cost	Total
count	4500.000000	4500.000000	4500.0	4.500000e+03	4.500000e+03	
mean	62.895333	125.790667	550.0	2.812480e+06	3.659588e+05	
std	65.818530	131.637061	0.0	2.768574e+06	3.633074e+05	
min	1.000000	2.000000	550.0	6.500000e+04	9.750000e+03	
25%	7.500000	15.000000	550.0	5.100000e+05	6.120000e+04	
50%	38.000000	76.000000	550.0	1.850000e+06	2.419200e+05	
75%	105.000000	210.000000	550.0	4.590000e+06	5.805000e+05	
max	250.000000	500.000000	550.0	1.050000e+07	1.350000e+06	

8 rows × 30 columns



```
In [6]: df.isnull().sum()
```

```
Out[6]: Customer_ID          0
        Customer_Name       0
        State              0
        City               0
        Customer_Type       0
        Property_Type       0
        Installation_Date   0
        Solar_Brand         0
        Solar_Capacity_KW   0
        Number_of_Panels    0
        Panel_Wattage       0
        Inverter_Brand      0
        Installation_Type   0
        System_Type         0
        Solar_Cost          0
        Installation_Cost    0
        Total_Project_Cost  0
        Government_Subsidy  0
        Net_Cost_After_Subsidy 0
        Monthly_Units_Generated 0
        Monthly_Units_Consumed 0
        Monthly_Units_Exported 0
        Monthly_Units_Saved 0
        Annual_Units_Generated 0
        Annual_Units_Consumed 0
        Annual_Units_Saved 0
        Electricity_Rate_Per_Unit 0
        Monthly_Bill_Before_Solar 0
        Monthly_Bill_After_Solar 0
        Monthly_Savings     0
        Annual_Savings      0
        ROI_Years           0
        Lifetime_Savings    0
        Annual_Maintenance_Cost 0
        Service_Warranty_Years 0
        Panel_Warranty_Years 0
        Performance_Warranty_Years 0
        Expected_Lifespan_Years 0
        CO2_Reduced_KG_Per_Year 0
        Trees_Equivalent    0
        Green_Energy_Percentage 0
        dtype: int64
```

```
In [7]: df.duplicated().sum()
```

```
Out[7]: 0
```

Step 3: Data Cleaning

```
In [8]: # Date Convert
df['Installation_Date'] = pd.to_datetime(df['Installation_Date'])
```

C:\Users\PIXEL-FC93\AppData\Local\Temp\ipykernel_19540\2573056259.py:2: UserWarning: Parsing dates in %d/%m/%Y format when dayfirst=False (the default) was specified. Pass `dayfirst=True` or specify a format to silence this warning.
df['Installation_Date'] = pd.to_datetime(df['Installation_Date'])

```
In [9]: df['Installation_Year'] = df['Installation_Date'].dt.year
df['Installation_Month'] = df['Installation_Date'].dt.month_name()
```

Step 4: Feature Engineering

```
In [10]: df['Actual_Payback_Years']=(df['Net_Cost_After_Subsidy']/df['Annual_Savings'])
```

```
In [11]: df['Actual_Payback_Years']
```

```
Out[11]: 0      3.613711
1      5.999023
2      3.837803
3      4.329917
4      4.472245
...
4495   3.543535
4496   3.152241
4497   3.619784
4498   3.741929
4499   3.030442
Name: Actual_Payback_Years, Length: 4500, dtype: float64
```

Step 5: Lifetime Profit

```
In [12]: df['Lifetime_Profit'] = (
    (df['Annual_Savings'] * df['Expected_Lifespan_Years'])
    -
    df['Net_Cost_After_Subsidy']
)
```

```
In [13]: # Subsidy %  
df['Subsidy_Percentage'] = (  
    df['Government_Subsidy']  
    /  
    df['Total_Project_Cost']  
    ) * 100
```

```
In [14]: # Bill Reduction %  
df['Bill_Reduction_Percentage'] = (  
    (  
    df['Monthly_Bill_Before_Solar']  
    -  
    df['Monthly_Bill_After_Solar']  
    )  
    /  
    df['Monthly_Bill_Before_Solar']  
    ) * 100
```

```
In [15]: # Solar Return %  
df['Solar_Return_Percentage'] = (  
    df['Annual_Savings']  
    /  
    df['Net_Cost_After_Subsidy']  
    ) * 100
```

```
In [ ]: # Executive KPIs
```

```
In [16]: Total_Customers = df['Customer_ID'].nunique()  
  
Total_Capacity = df['Solar_Capacity_KW'].sum()  
  
Total_Investment = df['Total_Project_Cost'].sum()  
  
Total_Subsidy = df['Government_Subsidy'].sum()  
  
Total_Annual_Savings = df['Annual_Savings'].sum()  
  
Total_CO2_Reduced = df['CO2_Reduced_KG_Per_Year'].sum()
```

Step 6: Adoption Analysis

```
In [17]: # State That Adopt Solar Most  
df.groupby('State')['Customer_ID'].count().sort_values(ascending=False)
```

```
Out[17]: State  
Gujarat          1152  
Maharashtra      488  
Uttar Pradesh    351  
Tamil Nadu       340  
Karnataka        334  
Uttarakhand      179  
Bihar            177  
Telangana        175  
Andhra Pradesh   170  
Haryana          169  
West Bengal      167  
Odisha           166  
Rajasthan        163  
Chandigarh       158  
Madhya Pradesh   158  
Chhattisgarh     153  
Name: Customer_ID, dtype: int64
```

☀️ Gujarat leads solar adoption with 1,152 installations, representing nearly 26% of all projects and positioning itself as a key contributor to India's renewable energy growth.

```
In [18]: # The Cities Which Lead  
df.groupby('City')['Customer_ID'].count().sort_values(ascending=False)
```

```
Out[18]: City  
Surat          573  
Ahmedabad      415  
Lucknow        182  
Dehradun       179  
Patna          177  
Chennai        175  
Hyderabad      175  
Mumbai         171  
Mysore         170  
Visakhapatnam  170  
Gurugram       169  
Noida          169  
Kolkata        167  
Nagpur         166  
Bhubaneswar    166  
Coimbatore     165  
Bangalore      164  
Vadodara       164  
Jaipur         163  
Indore         158  
Chandigarh     158  
Raipur         153  
Pune           151  
Name: Customer_ID, dtype: int64
```

 Surat recorded 573 solar installations, making it the most active city in the dataset and highlighting growing consumer interest in clean energy solutions.

```
In [19]: # Residential VS Industrial  
df['Customer_Type'].value_counts()
```

```
Out[19]: Customer_Type  
Commercial      1546  
Industrial       1494  
Residential     1460  
Name: count, dtype: int64
```

 Commercial customers account for 1,546 installations, slightly surpassing Industrial and Residential sectors, indicating strong business adoption of solar energy.

Step 7: Cost Analysis

```
In [20]: # Average Project Cost by State
df.groupby('State')['Total_Project_Cost'].mean().apply(lambda x: f"{x:,.0f}")
```

```
Out[20]: State
Andhra Pradesh    3,248,316
Bihar             3,146,922
Chandigarh        3,764,859
Chhattisgarh      3,604,028
Gujarat           2,931,964
Haryana           2,898,817
Karnataka          3,301,985
Madhya Pradesh    3,050,994
Maharashtra       3,361,859
Odisha            3,175,074
Rajasthan         2,752,874
Tamil Nadu        3,492,246
Telangana         2,750,504
Uttar Pradesh     3,021,233
Uttarakhand       3,354,469
West Bengal       3,886,695
Name: Total_Project_Cost, dtype: object
```

📌 West Bengal recorded the highest average project cost of ₹38.87 Lakhs, approximately 33% higher than Gujarat, indicating larger-scale installations and investments

```
In [21]: # Brand-Wise Cost
df.groupby('Solar_Brand')['Total_Project_Cost'].mean().apply(lambda x: f"{x:,.0f}")
```

```
Out[21]: Solar_Brand
Adani Solar       3,130,255
Goldi Solar       3,596,577
RenewSys          3,393,832
Tata Power Solar  3,108,730
Vikram Solar      3,122,922
Waaree            3,087,904
Name: Total_Project_Cost, dtype: object
```

```
In [22]: # Capacity-wise Cost
df.groupby('Solar_Capacity_KW')['Total_Project_Cost'].mean().apply(lambda x: f"{x:,.0f} ₹")
```

```
Out[22]: Solar_Capacity_KW
1.0      74,750
1.5     114,240
2.0     147,032
2.5     190,400
3.0     220,905
...
246.0    11,571,840
247.0    11,618,880
248.0    11,665,920
249.0    11,712,960
250.0    11,760,000
Name: Total_Project_Cost, Length: 247, dtype: object
```

Step 8: Subsidy Impact Analysis

```
In [23]: # Average Subsidy by Capacity
df.groupby('Solar_Capacity_KW')['Government_Subsidy'].mean()
```

```
Out[23]: Solar_Capacity_KW
1.0      40000.0
1.5      45000.0
2.0      40000.0
2.5      45000.0
3.0      78000.0
...
246.0         0.0
247.0         0.0
248.0         0.0
249.0         0.0
250.0         0.0
Name: Government_Subsidy, Length: 247, dtype: float64
```

```
In [24]: # Customer Type wise Susidy
df.groupby('Customer_Type')['Government_Subsidy'].mean()
```

```
Out[24]: Customer_Type
Commercial      0.000000
Industrial      0.000000
Residential    67360.958904
Name: Government_Subsidy, dtype: float64
```

 Residential customers receive an average subsidy of ₹67,360, significantly lowering installation costs and improving solar affordability for households.

```
In [25]: # Subsidy vs Payback
df[['Government_Subsidy', 'Actual_Payback_Years']].corr()
```

Out[25]:

	Government_Subsidy	Actual_Payback_Years
Government_Subsidy	1.000000	0.233819
Actual_Payback_Years	0.233819	1.000000

STEP 9: Electricity Generation Analysis

```
In [26]: # Brand-wise Generation
df.groupby('Solar_Brand')['Annual_Units_Generated'].mean()
```

Out[26]:

Solar_Brand	
Adani Solar	97390.323055
Goldi Solar	119813.961562
RenewSys	110069.238495
Tata Power Solar	96174.019443
Vikram Solar	97181.826795
Waaree	96124.076496
Name: Annual_Units_Generated, dtype: float64	

⚡ Goldi Solar generates an average of 119,814 units annually, enabling higher electricity savings and improved long-term financial returns.

```
In [27]: # Capacity Wise Generation
df.groupby('Solar_Capacity_KW')['Annual_Units_Generated'].mean()
```

Out[27]:

Solar_Capacity_KW	
1.0	1560.000000
1.5	2386.300900
2.0	3161.977101
2.5	3983.569626
3.0	4745.639716
...	
246.0	399182.376667
247.0	378466.340000
248.0	407236.280000
249.0	403469.420000
250.0	396338.522000
Name: Annual_Units_Generated, Length: 247, dtype: float64	

```
In [28]: # State-wise Generation
df.groupby('State')['Annual_Units_Generated'].mean()
```

```
Out[28]: State
Andhra Pradesh    105083.433824
Bihar             101373.696045
Chandigarh       122643.341962
Chhattisgarh     117312.921895
Gujarat          96640.944210
Haryana          84470.769231
Karnataka        102558.926168
Madhya Pradesh   89008.860759
Maharashtra      102238.188750
Odisha           92942.168675
Rajasthan        79827.975460
Tamil Nadu       108038.888059
Telangana        79898.742857
Uttar Pradesh    93228.975783
Uttarakhand      108578.888045
West Bengal      126336.965689
Name: Annual_Units_Generated, dtype: float64
```

STEP 10: Savings Analysis

```
In [29]: # brand-wise Saving
df.groupby('Solar_Brand')['Annual_Savings'].mean().apply(lambda x: f"{x:,.0f}")
```

```
Out[29]: Solar_Brand
Adani Solar      875,900
Goldi Solar      1,113,929
RenewSys         1,021,512
Tata Power Solar 872,035
Vikram Solar     874,888
Waaree           865,688
Name: Annual_Savings, dtype: object
```

```
In [30]: # State-wise Savings
df.groupby('State')['Annual_Savings'].mean().apply(lambda x: f"{x:,.0f}")
```

```
Out[30]: State
Andhra Pradesh    1,003,080
Bihar             951,411
Chandigarh       1,173,299
Chhattisgarh     1,091,936
Gujarat          858,886
Haryana          732,537
Karnataka        935,082
Madhya Pradesh   783,276
Maharashtra      916,364
Odisha           846,110
Rajasthan        715,436
Tamil Nadu       962,141
Telangana        707,700
Uttar Pradesh    853,016
Uttarakhand      1,033,914
West Bengal      1,170,449
Name: Annual_Savings, dtype: object
```

```
In [31]: # Customer-wise Saving
df.groupby('Customer_Type')['Annual_Savings'].mean().apply(lambda x: f"{x:,.0f}")
```

```
Out[31]: Customer_Type
Commercial      565,706
Industrial      2,078,087
Residential      66,174
Name: Annual_Savings, dtype: object
```

💰 Industrial customers save an average of ₹20.78 Lakhs annually, generating nearly four times the savings achieved by commercial installations.

STEP 11: Payback Analysis

```
In [32]: # Capacity-wise ROI
df.groupby('Solar_Capacity_KW')['Actual_Payback_Years'].mean()
```

```
Out[32]: Solar_Capacity_KW
1.0      2.522957
1.5      3.172237
2.0      3.877677
2.5      3.895632
3.0      3.533123
...
246.0    3.357421
247.0    3.932625
248.0    3.591436
249.0    2.547459
250.0    3.389227
Name: Actual_Payback_Years, Length: 247, dtype: float64
```

```
In [33]: # Brand-wise ROI
df.groupby('Solar_Brand')['Actual_Payback_Years'].mean()
```

```
Out[33]: Solar_Brand
Adani Solar      3.833875
Goldi Solar      3.494025
RenewSys         3.604819
Tata Power Solar 3.860583
Vikram Solar     3.852087
Waaree           3.811371
Name: Actual_Payback_Years, dtype: float64
```

 Goldi Solar achieves the fastest investment recovery at just 3.49 years, allowing customers to generate net savings sooner than competing brands.

```
In [34]: # Customer-wise ROI
df.groupby('Customer_Type')['Actual_Payback_Years'].mean()
```

```
Out[34]: Customer_Type
Commercial      3.918408
Industrial       3.488237
Residential      3.962759
Name: Actual_Payback_Years, dtype: float64
```

STEP 12: Lifetime Wealth Analysis

```
In [35]: # Brand Wise Lifetime Profit
df.groupby('Solar_Brand')['Lifetime_Profit'].mean().apply(lambda x: f"{x:,.0f}")
```

```
Out[35]: Solar_Brand
Adani Solar      18,788,496
Goldi Solar      24,272,319
RenewSys         22,167,706
Tata Power Solar 18,714,501
Vikram Solar     18,771,032
Waaree           18,576,200
Name: Lifetime_Profit, dtype: object
```

 Goldi Solar delivers an average lifetime profit of ₹2.42 Crores, transforming a one-time investment into a long-term wealth-generating asset.

```
In [36]: # Capacity-wise Lifetime Profit
df.groupby('Solar_Capacity_KW')['Lifetime_Profit'].mean().apply(lambda x: f"{x:,}.
```

```
Out[36]: Solar_Capacity_KW
1.0      314,339
1.5      484,143
2.0      597,791
2.5      805,063
3.0      891,444
...
246.0    75,658,118
247.0    62,243,242
248.0    69,561,823
249.0   103,234,909
250.0    76,456,223
Name: Lifetime_Profit, Length: 247, dtype: object
```

```
In [37]: # State-wise Lifetime Profit
df.groupby('State')['Lifetime_Profit'].mean().apply(lambda x: f"{x:,.0f}")
```

```
Out[37]: State
Andhra Pradesh    21,848,985
Bihar             20,664,011
Chandigarh        25,585,991
Chhattisgarh      23,715,236
Gujarat           18,563,365
Haryana           15,434,779
Karnataka         20,095,639
Madhya Pradesh    16,552,200
Maharashtra       19,569,541
Odisha            17,992,374
Rajasthan         15,154,906
Tamil Nadu        20,581,753
Telangana         14,969,287
Uttar Pradesh     18,326,995
Uttarakhand       22,516,976
West Bengal       25,392,860
Name: Lifetime_Profit, dtype: object
```

STEP 13: Solar vs Traditional Investment

```
In [38]: # Average Solar Return
df['Solar_Return_Percentage'].mean()
```

```
Out[38]: 27.357691548482613
```

 Solar installations generate returns exceeding 27%, outperforming Fixed Deposits and many traditional investment vehicles by a significant margin.

```
In [ ]: # FD = 6-7%
        # Savings Account = 3-4%
        # Solar Return = 27.35%
```

STEP 14: Environmental Impact

```
In [39]: # Total CO2 Reduced
df['CO2_Reduced_KG_Per_Year'].sum()
```

```
Out[39]: 375116419.45000005
```

 The analyzed installations reduce approximately 375 Million kg of CO₂ annually, contributing substantially to climate change mitigation efforts.

Step 15: Trees Equivalent

```
In [40]: df['Trees_Equivalent'].sum()
```

```
Out[40]: 17050746.700000003
```

 The environmental benefit is equivalent to planting over 17 Million trees, demonstrating the large-scale ecological impact of solar adoption.

Step 16: Investment Analysis

```
In [41]: def recommendation(row):

        if row['Actual_Payback_Years'] <= 5:
            return 'Excellent Investment'

        elif row['Actual_Payback_Years'] <= 7:
            return 'Good Investment'

        else:
            return 'Average Investment'

df['Recommendation'] = df.apply(recommendation, axis=1)
```

```
In [43]: df['Recommendation'].value_counts()
```

```
Out[43]: Recommendation
Excellent Investment    4210
Good Investment         287
Average Investment         3
Name: count, dtype: int64
```

Project Conclusion

1. 🌞 Gujarat emerged as the solar adoption leader with 1,152 installations, while Surat topped all cities with 573 active solar projects.
2. 💰 Government subsidies reduced installation costs by an average of ₹67,360 for residential customers, improving affordability and accelerating adoption.
3. ⚡ Top-performing systems generated over 1.19 Lakh units annually, creating substantial electricity savings for customers.
4. 🚀 Most installations recovered their investment within 3.5–5 years, making solar one of the fastest-paying infrastructure investments.
5. 💎 Several installations generated lifetime profits exceeding ₹2 Crores, proving solar's ability to create long-term financial value.
6. 📈 Average solar returns exceeded 27%, significantly outperforming traditional savings and fixed-income investment options.
7. 🌍 Collectively, the installations reduced over 375 Million kg of CO₂ emissions annually, equivalent to planting more than 17 Million trees.

🌞 A typical solar customer invests once, recovers the investment within 4 years, earns lakhs to crores in lifetime savings, and contributes to reducing 375 Million kg of carbon emissions annually. Solar is not just a power solution –it is a high-return financial asset and a long-term sustainability investment.

In []: Analysis By - Mr.Nirav Trivedi